

Product Requirement Document

Product Name

AI-Based Chatbot for Support

Problem Statement

Customers often experience delays and inconsistencies when seeking support, leading to frustration and increased operational costs for businesses. Existing support channels can be inefficient and struggle to handle a high volume of similar queries effectively.

Goals

- Improve customer satisfaction by providing instant and accurate answers to common queries.
- Reduce the workload on human support agents by automating responses to frequently asked questions.
- Decrease average support resolution time.
- Enhance the accuracy and relevance of chatbot responses over time through continuous learning.

User Personas

- End-users/Customers: Individuals seeking quick and accurate answers to their product/service-related questions.
- Customer Support Agents: Agents who need to offload routine queries and focus on complex issues.
- Administrators/Managers: Stakeholders responsible for managing and improving customer support operations.

Functional Requirements

- The chatbot shall accurately understand user queries using Natural Language Processing (NLP).
- The chatbot shall provide relevant and concise answers to user questions based on a knowledge base.
- The chatbot shall learn from user interactions and feedback to improve its response accuracy over time.

- The chatbot shall maintain context during a conversation.
- The chatbot shall be accessible via various platforms (e.g., web, mobile app).
- The chatbot shall have an administration interface for knowledge base management and performance monitoring.

Non Functional Requirements

- Performance: The chatbot shall respond to queries within 2 seconds.
- Scalability: The chatbot shall be able to handle up to 1000 concurrent users without degradation in performance.
- Security: The chatbot shall comply with industry-standard data privacy and security regulations (e.g., GDPR, CCPA).
- Reliability: The chatbot shall have an uptime of 99.9%.
- Usability: The chatbot interface shall be intuitive and easy for users to interact with.

Constraints

- Initial development budget limited to X USD.
- Development timeline of 6 months for the first release.
- Reliance on existing internal knowledge base for initial training data.

Assumptions

- Availability of sufficient and high-quality training data for NLP model development.
- Users will be willing to interact with an AI-based chatbot for support.
- Integration with existing customer support systems (if any) is feasible.
- The knowledge base will be regularly updated and maintained.

Out Of Scope

- Proactive outreach to customers.
- Multi-language support (initial release).

- Complex transactional capabilities (e.g., order placement, account changes, beyond simple query answering).
- Voice interaction (text-based only for initial release).
- Advanced sentiment analysis leading to proactive intervention.